

Ahmer Jamil

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Education

University of Florida

PhD in Computer Engineering

Area of study: Dependable Systems for HPC, Fault Tolerance and Machine Learning

Aug. 2025 – Present

Gainesville, Florida

Lahore University of Management Sciences

BSc in Computer Science

Graduate with Distinction, CGPA 3.78

August. 2019 – June. 2023

Lahore, Pakistan

Experience

University of Iowa – *Graduate Research Assistant*

Jan. 2025 – August. 2025

- LLVM/runtime tools for instruction-level resilience analysis; fault detection in HPC workloads
- Resilience tooling and analyses; supported experiments for dependable systems research

Systems Limited – *Associate Consultant , AI*

June. 2023 – July. 2024

- Developed LLM solutions with Microsoft using Azure and RAG, fine-tuning models with PEFT and LoRA
- Delivered ML solutions for international clients like Etisalat using PySpark, DataRobot, and Databricks

VentureDive – *Data Analysis Intern*

July. 2022 – August. 2022

- Utilized Selenium and online plotting tools to generate GeoJson data for the Constituencies of Pakistan
- Developed an interactive webpage by plotting the GeoJson file with Python, Flask, and HTML.

Habib Bank Limited – *Branch Banking Intern*

June. 2022 – July. 2022

- Developed ML models to measure employee effectiveness within branch banking operations
- Analyzed employee performance metrics to identify patterns for targeted training and performance improvement.

JS BANK – *Data Science Intern*

June. 2021 – August. 2021

- Leveraged customer data through SQL queries, and translated to predictive insights analysis models
- Converted insights into actionable strategies, enhancing informed decision-making across departments

Projects

LLM-Powered Chatbot

- Developed a knowledge-based chatbot using Retrieval-Augmented Generation (RAG) and ChromaDB
- Integrated OpenAI GPT-3.5 for conversational fluency, with HTML/CSS front end and Python backend

Customer Segmentation

- Engineered pipelines in Python with scikit-learn and XGBoost for behavioral and demographic clustering.
- Applied probability calibration and ensemble trees with SHAP explanations, increasing campaign response by ~12%

Fake News Detection

- Implemented misinformation detection pipeline with scikit-learn and NLP libraries (NLTK, spaCy)
- Applied frequent pattern mining (Apriori, FP-Growth) with Random Forest and SVM classifiers

Fraud Detection

- Developed credit card fraud detection models using ensemble learning and cloud-based platforms (Snowflake, Dataiku)
- Addressed class imbalance and resource constraints through data preprocessing, SMOTE, and robust evaluation

Knowledge Distillation Research [GitHub]

- Investigated compression methods for deep neural networks with teacher-student architectures implemented in PyTorch
- Showed direct pipelines reduced accuracy loss by ~5–7% compared to intermediary methods

Technical Skills

Programming Languages & Databases: Python, C++, SQL, R, Haskell, JavaScript, Oracle, MySQL

Libraries & Frameworks: NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, HuggingFace, OpenCV, PySpark, Flask, React Native, Spark, NLP

Tools & Platforms: AzureML, AWS, DataRobot, Dataiku, Docker, Linux, Tableau, Power BI

Generative AI / NLP: RAG, GPT-3/4, Vector DBs, LangChain, ChromaDB, AzureAI, Fine-tuning, Document Intelligence

Other Skills: Data Modeling, Software Architecture, Churn Analysis, A/B Testing, Bayesian Inference

Productivity & Design: Microsoft Office, Canva, Figma

Certifications: Generative AI Fundamentals (Databricks), Large Language Models: Application through Production (Databricks), Microsoft Azure SQL (Coursera), Intermediate Data Modeling in Power BI (DataCamp)